

Gerard Morales Riera

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Education

- **Universitat Politècnica de Catalunya (UPC)** Barcelona, Spain
Bachelor's degree in Engineering Physics. 2020 – 2025
 - **Universitat Politècnica de Catalunya (UPC)** Terrassa, Spain
Bachelor's degree in Aerospace Engineering. 2020 – 2025
- Joint degrees as part of the intensive CFIS-UPC program, after selection process.
- **La Salle Girona** Girona, Spain
High School, technological; 9.79/10 with honours mention (top 3%). 2018 – 2020

Work Experience

- **Own project** ($\approx 30\text{h/week}$) Barcelona, Spain
With a partner, building location-aware AI agents for user guidance. September 2025 – present
- **Project Lead** (part-time) at **CIMNE** Barcelona, Spain
Directing the development of a system to optimize flight operations for unmanned fire extinguishing airplanes. Supervising 4 degree theses too. September 2025 – present
- **Machine Learning Research Intern** (full-time) at **Fundació i2Cat** Barcelona, Spain
Developed time-series forecasting tools with ML for interurban road safety. May 2025 – Jul 2025
- **Student Researcher** (full-time) at **Texas A&M University** College Station, TX, US
Built and trained AI models for environmental monitoring satellites (with focus on climate data acquisition) using RL and transformers. Aug 2024 – May 2025
- **Software Engineering Intern** (full-time) at **Nostrum Biodiscovery** Barcelona, Spain
Added a multi-nucleotide loop closure algorithm to a protein folding solver. Jun 2023 – Sep 2023
- **Aerodynamics Engineer** ($\approx 20\text{h/week}$) at **UPC ecoRacing** Terrassa, Spain
Designed, manufactured and tested the aerodynamics of a competition car. Competed in the Netherlands, Germany, Hungary and Spain. Oct 2021 – Mar 2023

Volunteering

- **Study Resources Collection** with three classmates on our own Terrassa, Spain
Created and shared a Drive folder with studying material for Aerospace Eng. Jan 2024 – Mar 2024
- **Food Bank Volunteer Staff** (part-time) at **Banc dels Aliments Girona** Girona, Spain
Helped with storage and classification of food packages during the summer. Jul 2018 – Aug 2018

Awards

- Fundació Mir-Puig scholarship for studies abroad (degree thesis).
- High School 1st class honours, with university tuition scholarship.

Skills

- **Programming:** Python, C++, Matlab, R, Dart, HTML, SQL.
- **Frameworks & Libraries:** PyTorch, Cuda, Pandas, Numpy, Scikit-Learn, Neo4j, Catia V5, Ansys Fluent, Ansys Systems Tool Kit, Django, FastAPI, Vue, Flutter.
- **Artificial Intelligence:** deep learning, transformers, SL, SSL, RL (PPO, SAC), voice models.
- **Aerospace:** CAD design, CFD simulations, aerodynamics design and validation.
- **Computer Tools:** Git, Anaconda, Linux, Slurm, Google Cloud Run, LaTeX, Bash.
- **Languages:** English (C1 Advanced, 110/120 TOEFL), Spanish (Native), Catalan (Native).